A2 1 · IS01

Flat file

A2 1 · IS01

Data redundancy

A2 1 · IS01

Data inconsistency

A2 1 · IS01

Database management system (DBMS)

A2 1 · IS01

Data independence

A2 1 · IS01

Relational database

A2 1 · IS02

Entity

A2 1 · IS02

Attribute

Backs · print double-sided, flip on long edge

The unnecessary storage of the same data item more than once.

Data redundancy

A database held as a single table or file with no relationships to other tables.

Flat file

Software that allows a database to be defined, created, queried, updated and controlled, including security, integrity and concurrent access.

Database management system (DBMS)

A situation where copies of the same data item hold different values, so it is unclear which is correct.

Data inconsistency

A database in which data is stored in tables (relations) linked to each other through primary and foreign keys.

Relational database

The ability to change the way data is stored or structured without having to change the application programs that use it.

Data independence

A property or characteristic of an entity; it becomes a field in a table.

Attribute

An object, person, place or event about which data is stored; it becomes a table in a relational database.

Entity

A2 1 · IS02

Primary key

A2 1 · IS02

Foreign key

A2 1 · IS02

Composite key

A2 1 · IS02

Link entity

A2 1 · IS03

Normalisation

A2 1 · IS03

First normal form (1NF)

A2 1 · IS03

Second normal form (2NF)

A2 1 · IS03

Third normal form (3NF)

Backs · print double-sided, flip on long edge

An attribute in one table that is the primary key of another table and is used to link the two tables.

Foreign key

An attribute, or combination of attributes, that uniquely identifies each record in a table.

Primary key

An entity introduced to resolve a many-to-many relationship into two one-to-many relationships.

Link entity

A primary key made up of two or more attributes which together uniquely identify a record.

Composite key

A relation with no repeating groups in which every attribute holds a single atomic value.

First normal form (1NF)

A formal process of organising data into relations to minimise redundancy and prevent update, insertion and deletion anomalies.

Normalisation

A relation in 2NF with no transitive dependencies, so every non-key attribute depends only on the primary key.

Third normal form (3NF)

A relation in 1NF with no partial dependencies on a composite key.

Second normal form (2NF)

A2 1 · IS03

Partial dependency

A2 1 · IS03

Transitive dependency

A2 1 · IS03

Update anomaly

A2 1 · IS04

Data dictionary

A2 1 · IS04

Metadata

A2 1 · IS04

Entity integrity

A2 1 · IS04

Referential integrity

A2 1 · IS04

Index

Backs · print double-sided, flip on long edge

Where a non-key attribute depends on another non-key attribute rather than directly on the primary key.

Transitive dependency

Where a non-key attribute depends on only part of a composite primary key.

Partial dependency

A central store of metadata describing every table, field, data type, key, validation rule, index and relationship in a database.

Data dictionary

An inconsistency caused when a fact stored in several places is changed in some but not all of them.

Update anomaly

The rule that every table has a primary key whose values are unique and never null.

Entity integrity

Data that describes other data, such as a field's name, type and size.

Metadata

A sorted data structure of key values and pointers that lets the DBMS locate records quickly without scanning the whole table.

Index

The rule that every foreign key value must match an existing primary key value in the related table, preventing orphan records.

Referential integrity

A2 1 · IS04

Cascade delete

A2 1 · IS05

SQL

A2 1 · IS05

Data definition language (DDL)

A2 1 · IS05

Data manipulation language (DML)

A2 1 · IS05

Constraint

A2 1 · IS06

Aggregate function

A2 1 · IS06

GROUP BY

A2 1 · IS06

HAVING

Backs · print double-sided, flip on long edge

Structured Query Language: the standard declarative language for defining, querying and manipulating relational databases.

SQL

A referential integrity option in which deleting a parent record automatically deletes its related child records.

Cascade delete

The part of SQL used to retrieve and change data, e.g. SELECT, INSERT, UPDATE, DELETE.

Data manipulation language (DML)

The part of SQL used to create and change database structure, e.g. CREATE TABLE, ALTER TABLE, DROP TABLE.

Data definition language (DDL)

An SQL function that calculates a single value from a set of rows, such as COUNT, SUM, AVG, MIN or MAX.

Aggregate function

A rule defined on a table, such as PRIMARY KEY, FOREIGN KEY, NOT NULL or CHECK, that the DBMS enforces on every change.

Constraint

An SQL clause that filters groups after aggregation, in the way WHERE filters individual rows.

HAVING

An SQL clause that groups rows with the same value in a field so aggregate functions are calculated for each group.

GROUP BY

A2 1 · IS06

Join

A2 1 · IS06

Wildcard

A2 1 · IS07

Transaction

A2 1 · IS07

ACID

A2 1 · IS07

Record locking

A2 1 · IS07

Deadlock

A2 1 · IS07

Rollback

A2 1 · IS07

Distributed database

Backs · print double-sided, flip on long edge

A symbol used with LIKE to match unknown characters: % for any sequence and _ for exactly one character. Wildcard	An operation that combines rows from two or more tables by matching a foreign key with the related primary key. Join
Atomicity, consistency, isolation and durability: the properties that guarantee database transactions are processed reliably. ACID	A sequence of database operations treated as a single logical unit that must either complete entirely or not at all. Transaction
A situation where two or more transactions each hold a lock the other needs, so none can proceed. Deadlock	A concurrency control technique that prevents other users updating a record while one transaction is using it. Record locking
A single logical database whose data is stored at several sites linked by a network but appears to users as one database. Distributed database	Undoing all changes made by a transaction since it began, returning the database to its previous consistent state. Rollback

A2 1 · IS08

Big data

A2 1 · IS08

Data warehouse

A2 1 · IS08

ETL

A2 1 · IS08

Data mining

A2 1 · IS08

OLAP

A2 1 · IS09

Local area network (LAN)

A2 1 · IS09

Wide area network (WAN)

A2 1 · IS09

Topology

Backs · print double-sided, flip on long edge

A large store of integrated, historical, read-only data from many sources, organised for analysis and decision-making.

Data warehouse

Data sets so large, fast-moving or varied that they cannot be processed efficiently with traditional database tools.

Big data

The process of analysing large data sets to discover previously unknown patterns, relationships and trends.

Data mining

Extract, transform, load: the process of copying data from source systems, cleaning and converting it, and loading it into a data warehouse.

ETL

A network covering a single site or building, with infrastructure owned and managed by the organisation.

Local area network (LAN)

Online analytical processing: tools that let users analyse multidimensional data, for example by slicing and drilling down.

OLAP

The physical or logical arrangement of the devices and links in a network.

Topology

A network covering a large geographical area, connecting LANs using third-party telecommunications links.

Wide area network (WAN)

A2 1 · IS09

Switch

A2 1 · IS09

Router

A2 1 · IS09

Gateway

A2 1 · IS10

Protocol

A2 1 · IS10

OSI model

A2 1 · IS10

TCP/IP model

A2 1 · IS10

Encapsulation (networking)

A2 1 · IS10

TCP

Backs · print double-sided, flip on long edge

A device that connects different networks and forwards packets towards their destination using IP addresses and routing tables. Router	A device that connects devices within a LAN and forwards each frame only to the port of its destination, using MAC addresses. Switch
A set of rules that governs how data is formatted, transmitted and received so devices can communicate. Protocol	A device that connects two networks using different protocols and translates between them. Gateway
The four-layer model used by the internet: link, internet, transport and application. TCP/IP model	A seven-layer reference model of network communication: physical, data link, network, transport, session, presentation and application. OSI model
Transmission Control Protocol: a connection-oriented transport protocol giving reliable, ordered delivery using acknowledgements and retransmission. TCP	The process by which each layer adds its own header to data as it passes down the layers before transmission. Encapsulation (networking)

A2 1 · IS10

UDP

A2 1 · IS11

IP address

A2 1 · IS11

Subnet mask

A2 1 · IS11

DHCP

A2 1 · IS11

Network address translation (NAT)

A2 1 · IS11

Packet switching

A2 1 · IS11

Circuit switching

A2 1 · IS12

Bandwidth

Backs · print double-sided, flip on long edge

A logical numerical address that identifies a device on a network so data can be routed to it. IP address	User Datagram Protocol: a connectionless transport protocol that is fast but does not guarantee delivery or order. UDP
Dynamic Host Configuration Protocol: a service that automatically assigns IP addresses and network settings to devices. DHCP	A value that identifies which bits of an IP address form the network part and which form the host part. Subnet mask
A method of transmission in which data is split into packets that travel independently across shared links and are reassembled at the destination. Packet switching	A router technique that maps private IP addresses to a public IP address so many devices can share one public address. Network address translation (NAT)
The capacity of a communication channel (range of frequencies), which determines the maximum rate at which data can be transmitted. Bandwidth	A method of transmission in which a dedicated path is set up and reserved between sender and receiver for the whole communication. Circuit switching

A2 1 · IS12

Bit rate

A2 1 · IS12

Full-duplex

A2 1 · IS12

Asynchronous transmission

A2 1 · IS12

Parity bit

A2 1 · IS12

Cyclic redundancy check (CRC)

A2 1 · IS12

Automatic repeat request (ARQ)

A2 1 · IS13

Firewall

A2 1 · IS13

Proxy server

Backs · print double-sided, flip on long edge

A transmission mode in which data can travel in both directions at the same time. Full-duplex	The number of bits transmitted per second over a communication channel. Bit rate
An extra bit added to a group of bits so that the total number of 1s is even (even parity) or odd (odd parity), used to detect errors. Parity bit	Transmission without a shared clock, where each character is framed by start and stop bits. Asynchronous transmission
An error-correction method in which the receiver acknowledges each block and the sender retransmits any block that is negatively acknowledged or times out. Automatic repeat request (ARQ)	An error-detection method in which a remainder calculated by dividing the data by an agreed polynomial is sent and checked by the receiver. Cyclic redundancy check (CRC)
A server that acts as an intermediary between users and the internet, hiding internal addresses, filtering content and caching pages. Proxy server	Hardware or software that monitors network traffic and allows or blocks it according to a set of security rules. Firewall

A2 1 · IS13

Symmetric encryption

A2 1 · IS13

Asymmetric encryption

A2 1 · IS13

Digital signature

A2 1 · IS13

Digital certificate

A2 1 · IS13

Virtual private network (VPN)

A2 1 · IS14

Cloud computing

A2 1 · IS14

Infrastructure as a service (IaaS)

A2 1 · IS14

Platform as a service (PaaS)

Backs · print double-sided, flip on long edge

Encryption using a mathematically related key pair: data encrypted with the public key can only be decrypted with the private key, and vice versa. Asymmetric encryption	Encryption in which the same secret key is used both to encrypt and to decrypt data. Symmetric encryption
An electronic document issued by a certificate authority that binds a public key to the identity of its owner. Digital certificate	A hash of a message encrypted with the sender's private key, used to verify the sender's identity and that the message has not been altered. Digital signature
The on-demand delivery of computing resources such as servers, storage and software over the internet, usually paid for as used. Cloud computing	An encrypted connection across a public network that lets a remote user access a private network securely. Virtual private network (VPN)
A cloud model in which the provider supplies the hardware, operating system and development platform so customers only build and manage their own applications. Platform as a service (PaaS)	A cloud model in which the provider supplies virtualised hardware such as servers, storage and networking, and the customer manages the operating system and software. Infrastructure as a service (IaaS)

A2 1 · IS14

Software as a service (SaaS)

A2 1 · IS14

Hybrid cloud

A2 1 · IS14

Virtualisation

A2 1 · IS14

Hypervisor

A2 1 · IS15

Transaction processing system (TPS)

A2 1 · IS15

Management information system (MIS)

A2 1 · IS15

Decision support system (DSS)

A2 1 · IS15

Expert system

Backs · print double-sided, flip on long edge

A cloud deployment that combines a private cloud with public cloud services, allowing data and applications to move between them.

Hybrid cloud

A cloud model in which complete application software is hosted by the provider and accessed by users over the internet.

Software as a service (SaaS)

Software that creates, runs and manages virtual machines on a physical host computer.

Hypervisor

The use of software to create virtual versions of hardware, allowing several virtual machines to run on one physical computer.

Virtualisation

An information system that summarises data from transaction systems into regular, exception and on-demand reports for middle managers.

Management information system (MIS)

An information system that records and processes the routine day-to-day transactions of an organisation, in real time or in batches.

Transaction processing system (TPS)

A computer system that emulates the decision-making of a human expert in a specific domain using a knowledge base and an inference engine.

Expert system

An interactive information system that uses data and models to help managers make semi-structured and unstructured decisions, for example through what-if analysis.

Decision support system (DSS)

A2 1 · IS15

Knowledge base

A2 1 · IS15

Inference engine

A2 1 · IS16

Artificial intelligence (AI)

A2 1 · IS16

Machine learning

A2 1 · IS16

Supervised learning

A2 1 · IS16

Algorithmic bias

A2 1 · IS16

Internet of Things (IoT)

A2 1 · IS17

Personal data

Backs · print double-sided, flip on long edge

The part of an expert system that applies the rules in the knowledge base to the facts supplied in order to reach conclusions. Inference engine	The part of an expert system that stores facts and rules gathered from human experts. Knowledge base
A branch of AI in which algorithms improve their performance at a task by learning patterns from data rather than following explicitly programmed rules. Machine learning	The ability of computer systems to perform tasks that normally require human intelligence, such as recognising images, understanding language and making decisions. Artificial intelligence (AI)
Systematic unfairness in the outputs of an AI system, usually caused by biased or unrepresentative training data. Algorithmic bias	Machine learning in which a model is trained on labelled examples whose correct outputs are known. Supervised learning
Any information relating to an identified or identifiable living individual. Personal data	A network of everyday physical objects with embedded sensors, processors and connectivity that collect and exchange data. Internet of Things (IoT)

A2 1 · IS17

Data controller

A2 1 · IS17

Data processor

A2 1 · IS17

Data subject

A2 1 · IS17

Special category data

A2 1 · IS17

Computer Misuse Act 1990

A2 1 · IS17

Freedom of Information Act 2000

A2 1 · IS18

Digital divide

A2 1 · IS18

E-waste

Backs · print double-sided, flip on long edge

A person or organisation that processes personal data on behalf of a data controller.

Data processor

The person or organisation that decides the purposes for which, and the way in which, personal data is processed.

Data controller

Personal data needing extra protection, such as health, ethnicity, religion, political opinions, trade union membership, genetic and biometric data.

Special category data

The living individual whom personal data is about.

Data subject

UK law giving the public the right to request recorded information held by public authorities, subject to exemptions.

Freedom of Information Act 2000

UK law creating offences of unauthorised access to computer material, access with intent to commit further offences, and unauthorised acts that impair computers.

Computer Misuse Act 1990

Discarded electrical and electronic devices, which may contain toxic and valuable materials.

E-waste

The gap between people who have access to digital technology and the skills to use it, and those who do not.

Digital divide

A2 1 · IS18

Surveillance

A2 1 · IS18

Repetitive strain injury (RSI)

A2 1 · IS18

Sustainability

Backs · print double-sided, flip on long edge

Pain or damage to muscles, tendons or nerves caused by repeated movements such as typing or using a mouse.

Repetitive strain injury (RSI)

The monitoring of people's behaviour, activities or communications, for example by CCTV, tracking or interception.

Surveillance

Meeting present needs without compromising the ability of future generations to meet theirs, for example by reducing energy use and e-waste.

Sustainability